Med Fact Check - Verification Report

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Original Claim

Breast cancer can cause hearth attacks.

Final Verdict

SUPPORTED (99% confidence)

Medical Justification:

The original claim that breast cancer can cause heart attacks is supported by robust evidence showing a significantly elevated risk of cardiovascular events, including myocardial infarction, in breast cancer patients compared to non-cancer controls.

Aggregation Analysis:

The sole subclaim directly matches the original claim and is labeled 'supported' with high confidence (0.99). The justification confirms a statistically significant association between breast cancer and cardiovascular events, including myocardial infarction (heart attacks), with no refuting evidence.

Subclaim Analysis

Subclaim 1: Breast cancer can cause hearth attacks.

Verdict: SUPPORTED (99% confidence)

Justification: Breast cancer patients exhibit a significantly elevated risk of cardiovascular events compared to cancer-free controls, including myocardial infarction (MI) with a pooled hazard ratio of 1.98 (95% CI 1.24, 3.16). The evidence also notes increased risks for heart failure, atrial fibrillation, and cardiovascular death in breast cancer patients. No studies directly refute the association between breast cancer and cardiovascular events.

Evidence Sources:

Antiplatelet Therapy in Breast Cancer Patients Using Hormonal Therapy: Myths, Evidence and Potentialities - Systematic Review.

Systematic Review | 2018-Aug | URL: https://pubmed.ncbi.nlm.nih.gov/30183988/

"Breast cancer is the most frequently diagnosed tumor in women worldwide, with a significant impact on morbidity and mortality. Chemotherapy and hormone therapy have significantly reduced mortality; however, the adverse effects are significant. Aspirin has been incorporated into clinical practice for over 100 years at a low cost, making it particularly attractive as a potential agent in breast cancer prevention and as an adjunct treatment to endocrine therapy in the prophylaxis of cardiovascular complications. The objective of this study was to evaluate the role of aspirin in reducing the incidence of breast cancer and to evaluate the impact of its use on morbidity and mortality and reduction of cardiovascular events as adjuvant therapy during breast cancer treatment with selective estrogen receptor modulators. A systematic review was performed using the PRISMA methodology and PICO criteria, based on the MEDLINE, EMBASE and LILACS databases. The original articles of clinical trials, cohort, case-control studies and meta-analyses published from January 1998 to June 2017, were considered. Most studies showed an association between the use of selective estrogen receptor modulators and the increase in thromboembolic events. The studies suggest a protective effect of aspirin for cardiovascular events during its concomitant use with selective estrogen receptor modulators and in the prevention of breast cancer. This systematic review suggests that aspirin therapy combines the benefit of protection against cardiovascular events with the potential reduction in breast cancer risk, and that the evaluation of the benefits of the interaction of endocrine therapy with aspirin should be further investigated."

Cardiovascular outcomes in breast cancer survivors: a systematic review and meta-analysis.

Systematic Review | 2023-Dec-21 | URL: https://pubmed.ncbi.nlm.nih.gov/37499186/

"AIMS: It is unclear whether the future risk of cardiovascular events in breast cancer (Bc) survivors is greater than in the general population. This meta-analysis quantifies the risk of cardiovascular disease development in Bc patients, compared to the risk in a general matched cancer-free population, and reports the incidence of cardiovascular events in patients with Bc. METHODS AND RESULTS: We searched PubMed, Scopus, and Web of Science databases (up to 23 March 2022) for observational studies and post hoc analyses of randomized controlled trials. Cardiovascular death, heart failure (HF), atrial fibrillation (AF), coronary artery disease (CAD), myocardial infarction (MI), and stroke were the individual endpoints for our meta-analysis. We pooled incidence rates (IRs) and risk in hazard ratios (HRs), using random-effects meta-analyses. Heterogeneity was reported through the I2 statistic, and publication bias was examined using funnel plots and Egger's test in the meta-analysis of risk. One hundred and forty-two studies were identified in total, 26 (836 301 patients) relevant to the relative risk and 116 (2 111 882 patients) relevant to IRs. Compared to matched cancer-free controls, Bc patients had higher risk for cardiovascular death within 5 years of cancer diagnosis [HR = 1.09; 95% confidence interval (CI): 1.07, 1.11], HF within 10 years (HR = 1.21; 95% CI: 1.1, 1.33), and AF within 3 years (HR = 1.13; 95% CI: 1.05, 1.21). The pooled IR for cardiovascular death was 1.73 (95% CI 1.18, 2.53), 4.44 (95% CI 3.33, 5.92) for HF, 4.29 (95% CI 3.09, 5.94) for CAD, 1.98 (95% CI 1.24, 3.16) for MI, 4.33 (95% CI 2.97, 6.30) for stroke of any type, and 2.64 (95% CI 2.97, 6.30) for ischaemic stroke. CONCLUSION: Breast cancer exposure was associated with the increased risk for cardiovascular death, HF, and AF. The pooled incidence for cardiovascular endpoints varied depending on population characteristics and endpoint studied. REGISTRATION: CRD42022298741."

Impact of adjuvant trastuzumab therapy and its discontinuation on cardiac function and mortality in patients with early-stage breast cancer: An analysis based on the Japanese Receipt Claim Database.

Scientific Literature | 2024-12-31 | URL: https://doi.org/10.1016/j.breast.2024.103871

"Trastuzumab is an antibody drug against HER2 protein expressed on the surface of HER2-positive cancer cells. One year of trastuzumab for HER2-positive early breast cancer added to appropriate local treatment and chemotherapy reduces the risk of breast cancer recurrence to 60 % [1 , 2] and the survival rate even in patients older than 70 years [3]. However, Trastuzumab is cardiotoxic, which is often reversible but can cause cardiac decompensation [4]. Trastuzumab, followed by anthracycline, can cause heart failure, even in young patients, and should be treated with caution [5]. Trastuzumab-induced cardiac dysfunction may force discontinuation of adjuvant trastuzumab therapy, which may worsen breast cancer prognosis in these patients [6 , 7]. Although the completion of trastuzumab therapy is desirable to improve breast cancer prognosis, there are still unclear whether or how the treatment should be continued in the setting of reduced cardiac function. In a 2018 Finnish Real-World Data (RWD) analysis, 13 % of patients on trastuzumab had a reduced ejection fraction (EF), and 2.4 % discontinued treatment due to cardiotoxicity [8]."

Incident Myocardial Infarction, Heart Failure, and Oncologic Outcomes in Breast Cancer Survivors.

Scientific Literature | 2024-11-05 | URL: https://doi.org/10.1016/j.jaccao.2024.08.008

"It is well established that cardiotoxic cancer treatments and shared risk factors between cancer and CVD increase the prevalence of CVD in cancer survivors. 29 The potential reverse association (ie, CVD promoting cancer) has been a recent focus of investigation. There are several potential biological mechanisms through which MI and HF can promote carcinogenesis. 30 Koelwyn et al 31 showed that MI accelerates breast cancer progression in mice by innate immune reprogramming through epigenetic modifications of monocytes in the bone marrow. Myocardial remodeling in response to pressure overload has been shown to induce cell proliferation of breast and lung tumors. 32 Similarly, Meijers et al 33 showed that circulating factors secreted by the failing heart after an induced MI are able to promote tumor growth."

Aromatase inhibitors and risk of cardiovascular events in breast cancer patients: a systematic review and meta-analysis.

Systematic Review | 2019-Oct-29 | URL: https://pubmed.ncbi.nlm.nih.gov/31665091/

"BACKGROUND: Cardiovascular events (CVEs) was considered as one of the primary cause to reduce the quality of life in breast cancer patients with aromatase inhibitors (AIs) treatment, which has not been sufficiently addressed. The aim of this study was to assess the correlation between risk of CVEs and AIs in patients with breast cancer. METHODS: Included studies were obtained from the databases of Embase, Pubmed, Cochrane Library, Clinical Trials.gov, and reference lists. The main outcome measures were overall incidence, odds ratios (ORs), and 95% confidence intervals (CIs). Furthermore, the association and the risk differences among different tumor types, AIs,ages,or treatment regimens were conducted. Fixed-effect or random-effect models were applied in the statistical analyses according to the heterogeneity. Our analysis was performed according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement. RESULTS: Seventeen studies, which included 44,411 subjects, were included in our analyses. The overall incidence of CVEs in AIs group was 13.02% (95% CI: 8.15-20.17%) and almost all of the high-grade CVEs occurred in patients treated with AIs. The pooled ORs of CVEs was 0.9940 (95% CI: 0.8545-1.1562). Under sub-group analysis, the incidence of CVEs related to exemestane was higher than that of controls (OR?=?1.1564, 95% CI: 1.0656-1.2549), but no statistical differences in risk of CVEs were found in other sub-group analysis. No evidence of publication bias was found for incidence of CVEs in our meta-analysis by a funnel plot. CONCLUSIONS: These results suggest that patients with breast cancer treated with AIs do not have a significant risk of developing CVEs in comparison with the controls, and exemestane might not be considered as the alternative AI to the breast cancer patients from the perspective of CVEs. Further studies are recommended to investigate this association and the risk differences among different tumor types, AIs or treatment regimens."